VITATWIN: ENTERPRISE HEALTH ARCHITECTURE SCHEMATIC

PRIVACY-FIRST DESIGN · DPDP ACT 2023 ALIGNED

TIER 1: SECURE MULTI-SOURCE INGESTION & PRIVACY SANDBOX

Synthea FHIR EHR Baselines (Static)
- Demographics
- HbA1c
- Lipid Profiles

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Simulated Wearable IoT Telemetry (Dynamic)
- CGM
- RMSSD HRV
- Sleep Architecture

→

DPDP Act 2023 Zero-PII Compliance Gateway
(Data Anonymization & Tokenization)

TIER 2: THE IN-SILICO PREDICTIVE KINETICS ENGINE

Asynchronous Data Fusion Layer

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Google Gemini API: Physiological Inference Engine
(In-Silico Kinetics)

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Predictive Risk & Anomaly Detection
(2-Hour Postprandial Excursion Forecasting)

TIER 3: ACTIONABLE ROLE-BASED DUAL-PORTAL OUTPUTS

Patient Behavioral Portal (RBAC)
- What-If Simulation Engine (Localized Indian Glycemic Loads)
- Real-Time CGM Telemetry (WebSocket Live Data Feed)
- EHR Baseline Viewer (Read-Only Clinical History)

Clinician Command Portal (RBAC)
- Cohort Risk Triage Matrix (Automated Patient Prioritization)
- Twin Inspector Deep-Dive (Multi-System Metabolic Viewer)
- Pharmacotherapy Titration (Secure Clinical Intervention Feed)